Golay matroid with circuit-cocircuit intersection spectrum {0,2,4,6,8,12}

Houshan Fu

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Editorial note. One original exact-spectrum matroid construction thm:main, including its conjecture consequence. Complete original mathematical paper61–503, all polynomial code and support proofs, explicit realization sets and embedded bibliography retained verbatim. No external circuit enumeration or full Golay weight-table reconstruction.

Editorial note. Authors retain responsibility for mathematics. Collection and source positioning only; no proof repair, added assumptions, independent proof review, counterexample search or certification.

Allowed standard prerequisites (editorial).

- Binary matroid circuit/cocircuit spaces, polynomial binary codes and support/symmetric-difference identities. Native rank12/24-element/self-dual binary hypotheses and exact spectrum retained. General binary-representation support correspondence and standard code algebra remain original cited prerequisites.. All author-local polynomial code properties and support arguments included verbatim; no independent code enumeration, counterexample search or mathematical certification.

1 Introduction

All matroids considered are finite, and the matroid terminology follows Oxley [5]. For a matroid M, write $E(M)$, $r(M)$, and M^* for its ground set, rank, and dual, respectively. Denote its families of circuits and cocircuits by $\mathcal{C}(M)$ and $\mathcal{C}^*(M) = \mathcal{C}(M^*)$, respectively. A matroid M is *identically self-dual* if $M = M^*$. A matroid is *binary* if it is representable over the field GF(2) with two elements.

A *circuit-cocircuit intersection* is the intersection of a circuit and a cocircuit of the same matroid. No circuit-cocircuit intersection has size one [5, Proposition 2.1.11]. A matroid is binary if and only if every circuit-cocircuit intersection has even size [5, Theorem 9.1.2]. Oxley [4, Theorem 2.7] proved that a matroid with a circuit-cocircuit intersection of size at least four has one of size four. The following conjecture asks whether the intersection size can always be reduced by two.

Conjecture 1.1 ([5, Conjecture 15.9.5])**.** *Let M be a matroid with a k-element circuit-cocircuit intersection, where $k \geq 4$. Then M has a $(k-2)$-element circuit-cocircuit intersection.*

Recently, Shin proved the cases $k \leq 6$ for arbitrary matroids in [6, Theorem 2.2] and settled the case $k = 7$ in [7, Theorem 2.7]. Earlier, Kingan [2, Theorem 4.2] established Conjecture 1.1 for graphic matroids. Kingan and Lemos [3, Corollary 1.2] subsequently

proved a stronger result for regular matroids: every element of a k-element circuit-cocircuit intersection, where $k \geq 4$, belongs to one of size $k-2$.

The initial motivation for this work was to determine whether Conjecture 1.1 holds for the broader class of binary matroids. Within this class, Kingan [2, Theorem 2.3] proved that a matroid with a circuit-cocircuit intersection of size at least six has one of size six. Kingan and Lemos [3, pp. 472–473] used self-dual binary codes to construct matroids in their study of this conjecture. Their paper also contains a rank-nine example due to Oxley with circuit-cocircuit intersections of sizes ten and eight [3, pp. 473, 480].

The matroid represented by a generator matrix of the classical extended binary Golay code disproves the conjecture within the class of binary matroids: it has a twelve-element circuit-cocircuit intersection but none of size ten.

Theorem 1.2. *There is an identically self-dual binary matroid M of rank 12 on 24 elements such that*

$$\{|C \cap C^*| : C \in \mathcal{C}(M),\ C^* \in \mathcal{C}^*(M)\} = \{0, 2, 4, 6, 8, 12\}.$$

In particular, M has a 12-element circuit-cocircuit intersection and no 10-element circuit-cocircuit intersection.

The proof uses the code's duality and weight properties to characterize the circuits and cocircuits as supports of vectors of weight eight or twelve. If two such supports meet in ten elements, both have size twelve and their symmetric difference is the support of a vector of weight four. Such a vector contradicts the minimum weight eight. The required code properties are proved directly from the standard polynomial representation.

2 Construction and weight bounds

Let E be the set of column labels of a matrix A over a field $\mathbb{F}$. For each $e \in E$, write v_e for the column labelled by e. Set

$$\mathcal{I} = \{I \subseteq E : (v_e)_{e \in I} \text{ is linearly independent over } \mathbb{F}\}.$$

Then $M[A] = (E, \mathcal{I})$ is a matroid, called the *vector matroid* of A [5, Proposition 1.1.1]. The *row space* $R(A)$ is the subspace spanned over $\mathbb{F}$ by the rows of A. For each positive integer n, write $V(n, 2) = \mathrm{GF}(2)^n$. The *symmetric difference* of sets X and Y is $X \triangle Y = (X - Y) \cup (Y - X)$.

For a vector $u = (u_e)_{e \in E}$ over $\mathrm{GF}(2)$, its *support* is $\mathrm{supp}(u) = \{e \in E : u_e \neq 0\}$, and its *weight* is $\mathrm{wt}(u) = |\mathrm{supp}(u)|$. The *incidence vector* of $X \subseteq E$ is the vector with e-coordinate one for $e \in X$ and zero for $e \in E - X$. Orthogonality is taken with respect to the usual dot product over $\mathrm{GF}(2)$.

The construction follows the polynomial representation of the extended binary Golay code in Huffman and Pless [1, Section 10.1.2]. The binary Golay code of length 23 has generator polynomial

$$g(x) = 1 + x + x^5 + x^6 + x^7 + x^9 + x^{11} \in \mathrm{GF}(2)[x]. \tag{2.1}$$

The reciprocal polynomial of g is

$$h(x) = x^{11} g(x^{-1}) = 1 + x^2 + x^4 + x^5 + x^6 + x^{10} + x^{11}.$$

Direct multiplication in $\mathrm{GF}(2)[x]$ gives

$$g(x)h(x) = 1 + x + x^2 + \cdots + x^{22}. \tag{2.2}$$

For $0 \le i \le 11$ and $0 \le j \le 22$, define $a_{ij} \in \mathrm{GF}(2)$ by

$$x^i g(x) = \sum_{j=0}^{22} a_{ij} x^j. \tag{2.3}$$

For $0 \le i \le 11$, put $u_i = (a_{i0}, a_{i1}, \ldots, a_{i22}, 1) \in V(24, 2)$. Let A be the 12×24 matrix over $\mathrm{GF}(2)$ with rows $u_0, u_1, \ldots, u_{11}$ and columns labelled $0, 1, \ldots, 23$ in this order. Set $E = \{0, 1, \ldots, 23\}$ and $M = M[A]$. The matrix A and the matroid M are fixed for the remainder of the paper. For $0 \le i \le 11$, the polynomial $x^i g(x)$ has seven nonzero coefficients. The appended entry one is the overall parity-check coordinate, and $\mathrm{wt}(u_i) = 8$. The resulting row space $R(A)$ is the extended binary Golay code, with generator matrix A and associated matroid $M[A]$. The *orthogonal subspace* of $R(A)$ is

$$R(A)^\perp = \{v \in V(24, 2) : u \cdot v = 0 \text{ for every } u \in R(A)\}.$$

The next lemma uses the dot products of the rows to control the weights of their sums.

Lemma 2.1. *The row space $R(A)$ has dimension 12 and satisfies*

$$R(A) = R(A)^\perp.$$

Moreover, every vector in $R(A)$ has weight divisible by four.

Proof. We first show that $u_0, \ldots, u_{11}$ form a basis of $R(A)$. They span $R(A)$ by definition. Suppose $\sum_{i=0}^{11} c_i u_i = 0$, where $c_i \in \mathrm{GF}(2)$. In the first 23 coordinates, this relation reads $\sum_{i=0}^{11} c_i a_{ij} = 0$ for $0 \le j \le 22$. Multiplication by x^j and summation over $0 \le j \le 22$, together with (2.3), transform these coordinate relations into

$$0 = \sum_{j=0}^{22} \Big(\sum_{i=0}^{11} c_i a_{ij} \Big) x^j = \sum_{i=0}^{11} c_i x^i g(x) = g(x) \sum_{i=0}^{11} c_i x^i.$$

Since $g(x) \neq 0$ and $\mathrm{GF}(2)[x]$ is an integral domain, $\sum_{i=0}^{11} c_i x^i = 0$. Comparing coefficients shows that $c_i = 0$ for every i. Thus $u_0, \ldots, u_{11}$ form a basis of $R(A)$, and $\dim R(A) = 12$.

To prove orthogonality of the rows, substitute $h(x) = x^{11} g(x^{-1})$ into (2.2) and divide by x^{11}. The resulting identity in the Laurent polynomial ring $\mathrm{GF}(2)[x, x^{-1}]$ is

$$g(x) g(x^{-1}) = \sum_{d=-11}^{11} x^d.$$

For $0 \le i, j \le 11$, multiply (2.3) for i by the same identity for j with x replaced by x^{-1}:

$$\sum_{k,l=0}^{22} a_{ik} a_{jl} x^{k-l} = x^{i-j} g(x) g(x^{-1}) = \sum_{d=-11}^{11} x^{d+i-j}.$$

On the left, the constant terms are precisely those with $k = l$. On the right, only $d = j - i$ contributes a constant term, and $-11 \le j - i \le 11$. Hence $\sum_{k=0}^{22} a_{ik} a_{jk} = 1$. Since the last coordinate of each row is one, $u_i \cdot u_j = 1 + 1 = 0$. By bilinearity, any two linear combinations of the rows are orthogonal. Thus

$$R(A) \subseteq R(A)^\perp. \tag{2.4}$$

Since $\dim R(A)^\perp = 24 - \dim R(A) = 12$, the inclusion (2.4) is an equality.

To prove the divisibility assertion, consider $u, v \in V(24, 2)$. Binary addition corresponds to symmetric difference of supports:

$$\operatorname{supp}(u + v) = \operatorname{supp}(u) \triangle \operatorname{supp}(v). \tag{2.5}$$

Consequently, the weights satisfy

$$\operatorname{wt}(u + v) = \operatorname{wt}(u) + \operatorname{wt}(v) - 2|\operatorname{supp}(u) \cap \operatorname{supp}(v)|. \tag{2.6}$$

For $u, v \in R(A)$, (2.4) ensures $u \cdot v = 0$. In GF(2),

$$0 = u \cdot v = \sum_{e \in E} u_e v_e = \sum_{e \in \operatorname{supp}(u) \cap \operatorname{supp}(v)} 1.$$

Thus $|\operatorname{supp}(u) \cap \operatorname{supp}(v)|$ is even, and (2.6) implies

$$\operatorname{wt}(u + v) \equiv \operatorname{wt}(u) + \operatorname{wt}(v) \pmod 4. \tag{2.7}$$

For an arbitrary $u \in R(A)$, the basis coefficients lie in GF(2), so $u = \sum_{i \in I} u_i$ for some $I \subseteq \{0, \dots, 11\}$. Every partial sum belongs to $R(A)$. Repeated application of (2.7), using $\operatorname{wt}(u_i) = 8$, gives

$$\operatorname{wt}(u) \equiv \sum_{i \in I} \operatorname{wt}(u_i) = 8|I| \equiv 0 \pmod 4.$$

The empty sum corresponds to the zero vector and has weight zero. □

The divisibility by four reduces the minimum-weight bound to the exclusion of weight four. For this step, the proof uses the consecutive-root argument for cyclic codes described by Huffman and Pless [1, Theorem 4.5.3 and Example 4.5.5], including its application to the binary Golay code.

Lemma 2.2. *Every nonzero vector in $R(A)$ has weight at least eight.*

Proof. By Lemma 2.1, a nonzero vector in $R(A)$ can have weight less than eight only if its weight is four. Suppose, for a contradiction, that $u \in R(A)$ has weight four. Since $u_0, u_1, \dots, u_{11}$ form a basis of $R(A)$, write $u = \sum_{i=0}^{11} c_i u_i$ with $c_i \in \mathrm{GF}(2)$. Let $f(x)$ be the polynomial whose coefficients are the first 23 coordinates of u. For $0 \le j \le 22$, the coefficient of x^j in f is $\sum_{i=0}^{11} c_i a_{ij}$. Substitution from (2.3) expresses f as a multiple of g:

$$f(x) = \sum_{j=0}^{22} \Big(\sum_{i=0}^{11} c_i a_{ij} \Big) x^j = \sum_{i=0}^{11} c_i \Big(\sum_{j=0}^{22} a_{ij} x^j \Big) = \Big(\sum_{i=0}^{11} c_i x^i \Big) g(x). \tag{2.8}$$

Deleting the last coordinate of u removes either zero or one nonzero entry. Since $\operatorname{wt}(u) = 4$, the polynomial f has $t \in \{3, 4\}$ nonzero coefficients. Denote their positions by $0 \le e_1 < \cdots < e_t \le 22$. Since each nonzero coefficient equals one in GF(2), the expansion is

$$f(x) = \sum_{l=1}^{t} x^{e_l}. \tag{2.9}$$

Since g divides f by (2.8), the roots of g impose linear relations on the t nonzero coefficients of f. The contradiction follows once t of these relations form a nonsingular homogeneous system.

Let α be a root of g in an algebraic closure of GF(2). Multiplying (2.2) by $x - 1$ gives $(x - 1)g(x)h(x) = x^{23} - 1$. At $x = \alpha$, the left-hand side vanishes, so $\alpha^{23} = 1$. The equality

$g(1) = 1$ excludes $\alpha = 1$. As 23 is prime, the multiplicative order of α is 23. In characteristic two, $(a+b)^{2^j} = a^{2^j} + b^{2^j}$ for all elements a, b of the algebraic closure and every integer $j \geq 0$. Since each coefficient of g is zero or one, this identity implies $g(\alpha^{2^j}) = g(\alpha)^{2^j} = 0$ for every integer $j \geq 0$. In particular, α, α^2, and α^4 are roots of g. Since $2^8 \equiv 3 \pmod{23}$ and $\alpha^{23} = 1$, $\alpha^{2^8} = \alpha^3$. Thus, $\alpha, \alpha^2, \alpha^3, \alpha^4$ are roots of both g and f. For $1 \leq j \leq t \leq 4$, evaluation of (2.9) at α^j gives

$$0 = f(\alpha^j) = \sum_{l=1}^{t} \alpha^{je_l} \qquad (1 \leq j \leq t).$$

Equivalently,

$$\left(\alpha^{je_l}\right)_{1\leq j,l\leq t} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{2.10}$$

Since $\alpha^{23} = 1$, both $\alpha \neq 0$ and $\alpha^{e_l} \neq 0$ for $1 \leq l \leq t$. For $l, m \in \{1, \ldots, t\}$, if $\alpha^{e_l} = \alpha^{e_m}$, then $23 \mid (e_l - e_m)$ because α has multiplicative order 23. The bound $|e_l - e_m| < 23$ forces $e_l = e_m$, and the strict ordering of the exponents implies $l = m$. Thus, $\alpha^{e_1}, \ldots, \alpha^{e_t}$ are pairwise distinct. After α^{e_l} is factored from each column l, the remaining matrix is a Vandermonde matrix. The original determinant factors as

$$\det\left(\alpha^{je_l}\right)_{1\leq j,l\leq t} = \left(\prod_{l=1}^{t} \alpha^{e_l}\right) \det\left(\alpha^{(j-1)e_l}\right)_{1\leq j,l\leq t} = \left(\prod_{l=1}^{t} \alpha^{e_l}\right) \prod_{1\leq l<m\leq t} (\alpha^{e_m} - \alpha^{e_l}) \neq 0.$$

The coefficient matrix is nonsingular. Left multiplication by its inverse reduces (2.10) to $(1, \ldots, 1)^T = 0$, a contradiction since $t \geq 1$. □

3 Circuit-cocircuit intersections

The *circuit space* and *cocircuit space* of M are the subspaces of $V(24, 2)$ spanned over GF(2) by the incidence vectors of its circuits and cocircuits, respectively. For the binary representation A, these spaces are $R(A)^\perp$ and $R(A)$, respectively [5, Proposition 9.2.2]. The cocircuits are the inclusion-minimal nonempty supports of vectors in $R(A)$ [5, Proposition 9.2.4]. By duality, the circuits are the inclusion-minimal nonempty supports of vectors in $R(A)^\perp$. The rank bound on circuit size, together with the weight constraints in Lemma 2.1 and Lemma 2.2, determines these minimal supports.

Proposition 3.1. *The matroid M is identically self-dual, has rank 12, and satisfies*

$$\mathcal{C}(M) = \mathcal{C}^*(M) = \{\mathrm{supp}(u) : u \in R(A),\ \mathrm{wt}(u) \in \{8, 12\}\}.$$

Proof. By Lemma 2.1, the circuit and cocircuit spaces of M both equal $R(A)$. Equality of their inclusion-minimal nonempty supports implies $\mathcal{C}(M) = \mathcal{C}^*(M) = \mathcal{C}(M^*)$. Since the circuits determine a matroid, $M = M^*$. Moreover, $r(M) = \operatorname{rank} A = \dim R(A) = 12$.

For a circuit C and $e \in C$, the set $C - \{e\}$ is independent. Hence $|C| \leq r(M) + 1 = 13$. Since the incidence vector of C belongs to $R(A)$, Lemma 2.1 and Lemma 2.2 imply $8 \leq |C| \leq 13$ and $4 \mid |C|$. These constraints restrict $|C|$ to 8 or 12.

Conversely, let $u \in R(A)$ have weight eight or twelve. It remains to show that $\mathrm{supp}(u)$ is inclusion-minimal among the nonempty supports in $R(A)$. Suppose that $0 \neq v \in R(A)$ has $\mathrm{supp}(v) \subsetneq \mathrm{supp}(u)$. By (2.5), $\mathrm{supp}(u + v) = \mathrm{supp}(u) - \mathrm{supp}(v)$. The inclusion is proper, so

$u+v \neq 0$. By Lemma 2.2, the nonzero vectors v and $u+v$ satisfy $\mathrm{wt}(v) \geq 8$ and $\mathrm{wt}(u+v) \geq 8$. Since their supports partition $\mathrm{supp}(u)$,

$$\mathrm{wt}(u) = \mathrm{wt}(v) + \mathrm{wt}(u+v) \geq 16,$$

contrary to $\mathrm{wt}(u) \in \{8, 12\}$. Thus $\mathrm{supp}(u)$ is both a circuit and a cocircuit. □

The characterization in Proposition 3.1 reduces the intersection problem to pairs of supports of vectors of weight eight or twelve. The proof first excludes size ten and then exhibits pairs for all remaining possibilities.

Proof of Theorem 1.2. Let $C \in \mathcal{C}(M)$ and $C^* \in \mathcal{C}^*(M)$, and let u, v be their incidence vectors, respectively. Write $s = |C \cap C^*|$. By Proposition 3.1, $u, v \in R(A)$ and $|C|, |C^*| \in \{8, 12\}$. By (2.4), these incidence vectors satisfy $0 = u \cdot v = \sum_{e \in C \cap C^*} 1$ in GF(2). Thus, s is even. Together with the bound $s \leq \min\{|C|, |C^*|\} \leq 12$, this parity condition implies $s \in \{0, 2, 4, 6, 8, 10, 12\}$.

If $s = 10$, then $|C| = |C^*| = 12$. The vector $u + v$ lies in $R(A)$ and has support $C \triangle C^*$ by (2.5). Therefore,

$$\mathrm{wt}(u+v) = |C \triangle C^*| = |C| + |C^*| - 2s = 4.$$

The nonzero vector $u + v$ contradicts the weight bound in Lemma 2.2, excluding $s = 10$.

To realize the six remaining sizes, put $C_i = \mathrm{supp}(u_i)$ for $i \in \{0, 1, 3\}$. From (2.1) and (2.3),

$$C_i = \{i, i+1, i+5, i+6, i+7, i+9, i+11, 23\} \qquad (i \in \{0, 1, 3\}).$$

Since each $u_i \in R(A)$ has weight eight, Proposition 3.1 identifies C_0, C_1, C_3 as both circuits and cocircuits. Inspection of these sets gives

$$C_0 \cap C_3 = \{9, 23\}, \qquad C_0 \cap C_1 = \{1, 6, 7, 23\}.$$

Set $C' = \mathrm{supp}(u_0 + u_3) = C_0 \triangle C_3$ by (2.5). The values $|C_0| = |C_3| = 8$ and $|C_0 \cap C_3| = 2$ determine

$$|C'| = |C_0| + |C_3| - 2|C_0 \cap C_3| = 12, \qquad |C_0 \cap C'| = |C_0 - C_3| = 8 - 2 = 6.$$

Since $u_0 + u_3 \in R(A)$ has weight $|C'| = 12$, the set C' is also both a circuit and a cocircuit by Proposition 3.1. Together with $|C_0 \cap C_0| = 8$ and $|C' \cap C'| = 12$, the intersections above realize $s = 2, 4, 6, 8, 12$.

To realize $s = 0$, we use the complementary sets C' and $E - C'$. It remains to show that $E - C'$ is a cocircuit. Consider the all-one vector $\mathbf{1} = (1, \ldots, 1) \in V(24, 2)$. The even weights in $R(A)$ imply $\mathbf{1} \cdot u = \sum_{e \in E} u_e = 0$ in GF(2) for every $u \in R(A)$. By Lemma 2.1, $\mathbf{1} \in R(A)^\perp = R(A)$. By (2.5), the vector $\mathbf{1} + u_0 + u_3 \in R(A)$ has support

$$\mathrm{supp}(\mathbf{1} + u_0 + u_3) = E \triangle C' = E - C'.$$

Since $|E - C'| = 24 - |C'| = 12$, Proposition 3.1 identifies $E - C'$ as a cocircuit. The disjoint circuit-cocircuit pair $C', E - C'$ realizes $s = 0$. □

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